

PROJECT TOPIC: MEDICINAL PLANTS SURVEY & EXPERT INTERVIEW

Topic Description: Survey of Five Indigenous Medicinal Plants and Expert Interview with a Traditional Herbal Practitioner (Vaidu) in Maharashtra

Traditional Indian systems of medicine—including Ayurveda, Siddha, and folk ethnobotany—represent millennia of empirical botanical research and deep ecological wisdom. Long before synthetic chemical pharmaceuticals were manufactured, indigenous communities across Maharashtra relied exclusively on wild and cultivated medicinal herbs to cure common ailments, prevent epidemic diseases, and maintain family health.

This project presents a structured ethnobotanical field survey and formal expert interview investigating **Five Indigenous Medicinal Plants** found in our local surroundings. The study documents a detailed interaction with **Vaidu Shri Ramdas Patil**, an experienced traditional herbal practitioner and village elder possessing over 35 years of oral knowledge in identifying, harvesting, and preparing herbal remedies.

The project investigates five versatile healing plants: **Adulsa** (*Justicia adhatoda*), **Neem** (*Azadirachta indica*), **Tulsi** (*Ocimum sanctum*), **Aloe Vera / Korphad** (*Aloe barbadensis*), and **Bhui-Neem / Kalmegh** (*Andrographis paniculata*). By recording their active botanical parts, preparation methods, and current conservation threats, the study highlights the vital importance of preserving traditional ethnobotanical knowledge.

2. Importance of the topic (विषयाचे महत्त्व)

Documenting indigenous medicinal plants and interviewing traditional herbal healers provides critical insights into bio-resource conservation:

Why Ethnobotanical Documentation is Vital :-

- **Preventing the Extinction of Oral Indigenous Knowledge:** Traditional medicinal wisdom is passed down orally from generation to generation. With rapid urbanization and rural youth migration, this irreplaceable empirical knowledge is at risk of vanishing within a single generation if not scientifically recorded.
- **Affordable & Chemical-Free Primary Healthcare:** Indigenous medicinal herbs provide low-cost, accessible, and side-effect-free primary healthcare for minor ailments (coughs, fevers, skin infections, digestive disorders) for rural and urban households alike.
- **Defense Against Corporate Biopiracy:** Documenting local herbal applications in written records (such as People's Biodiversity Registers) provides legal evidence of prior art, preventing foreign pharmaceutical corporations from patenting traditional Indian knowledge.
- **Sustainable In-Situ Bio-Conservation:** Traditional harvesting ethics emphasize harvesting only leaves or bark without uprooting the whole plant, ensuring sustainable wild plant population survival.

Medicinal plants demonstrate that human health is directly reliant on the health of our botanical biodiversity.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The field project was guided by structured ethnobotanical, interview, and conservation objectives:

Structured Project Objectives :-

- **Expert Interview Execution:** To conduct a formal, structured interview with a traditional herbal practitioner (Vaidu / elder) regarding wild plant identification, harvesting seasons, and preparation.
- **Taxonomic & Morphological Documentation:** To identify, photograph, and describe the botanical features of 5 prominent medicinal plant species in the local area.
- **Therapeutic Applications Cataloging:** To document the specific plant parts used (leaf, root, bark, gel) and their traditional formulations for treating respiratory, digestive, and cutaneous disorders.
- **Current Ecological Status & Threat Audit:** To evaluate the abundance or depletion of wild medicinal herb populations due to concrete paving, over-grazing, and weedicide spraying.
- **Action Plan for Campus Herbal Gardens:** To design a practical protocol for establishing a living medicinal plant garden on junior college campuses.

The scope encompasses oral transcript recordings, botanical field sketches, and Ayurvedic reference cross-checks.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized ethnobotanical survey, expert interview, and specimen verification methodology was deployed:

Methodological Survey Protocol :-

- **Phase 1 – Practitioner Identification & Consent:** Contacted a recognized village herbal practitioner (Vaidu Shri Ramdas Patil), explained the educational purpose of the study, and scheduled an in-person interview.
- **Phase 2 – Structured Interview & Field Walk:** Conducted a 60-minute interview followed by a guided botanical field walk across local scrublands and home gardens to examine living specimens.
- **Phase 3 – Data Recording & Formulation Verification:** Documented local Marathi names, parts used, decoction/powder preparation techniques, and ethical wild-harvesting rules.
- **Phase 4 – Botanical & Pharmacognostic Verification:** Cross-verified all vernacular names with scientific binomial nomenclature and active biochemical alkaloids using Ayurvedic pharmacopeias.
- **Phase 5 – Report Synthesis:** Structured observational data into comparative tables and aligned results with Chapter 7 EVS competencies.

All herbal remedies recorded were documented for academic research and educational preservation purposes.

5. Observations (निरीक्षणे)

Field interviews and botanical surveys yielded comprehensive documentation of five prominent indigenous medicinal plants:

5.1 Tabular Record of 5 Medicinal Plants from Vaidu Interview

Medicinal Plant	Scientific Name & Family	Parts Used & Form	Traditional Therapeutic Uses & Active Biochemicals
Adulsa (अडुळसा)	Justicia adhatoda (Acanthaceae)	Fresh leaves; boiled decoction (Kadha) with honey.	Contains vasicine alkaloid; powerful bronchodilator and expectorant; cures chronic bronchitis, coughs, and respiratory congestion.
Neem (कडुनिंब)	Azadirachta indica (Meliaceae)	Leaves, bark, seed oil; paste or decoction.	Contains azadirachtin and nimbin; potent antiseptic, anti-fungal, and blood purifier; cures skin eczema, wounds, and controls blood sugar.
Tulsi (तुळस)	Ocimum sanctum (Lamiaceae)	Leaves and seeds; raw juice or herbal tea.	Contains eugenol and ursolic acid; natural adaptogen, antiviral, and anti-stress agent; cures common cold, malaria fevers, and boosts immunity.
Korphad / Aloe Vera	Aloe barbadensis (Asphodelaceae)	Inner mucilaginous leaf pulp/gel; fresh.	Rich in aloin and acemannan; soothing anti-inflammatory; treats thermal burns, cuts, skin hydration, and digestive ulcers.
Bhui-Neem / Kalmegh	Andrographis paniculata (Acanthaceae)	Whole aerial plant; extremely bitter powder/decoction.	Contains andrographolide ; known as the 'King of Bitters'; exceptional hepatoprotective agent curing liver disorders, jaundice, and viral fevers.

5.2 Expert Practitioner Insights Recorded

6. Analysis of data (माहितीचे विश्लेषण)

Analytical examination of the ethnobotanical interview reveals profound synergy between folk medicine and modern pharmacology:

Core Analytical Inferences :-

- **Scientific Validation of Folk Wisdom:** Modern biochemistry proves that the herbal preparations used by traditional Vaidus for centuries are backed by active pharmacological compounds. The vasicine in Adulsa is the chemical precursor for modern allopathic cough syrups (Bromhexine), confirming the scientific validity of folk ethnobotany.
- **The Destruction of Wild Herbal Nurseries:** The interview highlighted that traditional healers historically gathered wild herbs from unpaved road verges, seasonal streams, and sacred groves. Today, municipal paver blocks, concrete drains, and chemical weedicides have eliminated these wild micro-habitats, forcing Vaidus to travel 30 km to find wild herbs.
- **Commercial Over-Exploitation vs. Sustainable Use:** Unlike traditional healers who harvest only what is needed for immediate healing, commercial herbal cosmetic companies clear-fell wild medicinal plants recklessly, driving wild Sarpagandha and Kalmegh populations onto state red lists.
- **The Urgency of Domestic Cultivation:** Since wild habitats are shrinking, the only viable method to conserve medicinal biodiversity and protect family health is domestic backyard and college campus cultivation.

The analysis proves that traditional herbal knowledge is an invaluable scientific heritage that must be institutionalized.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Indigenous medicinal plants and the traditional healers who understand their secrets are vital components of India's biological and cultural wealth. Protecting these healing plants requires ending destructive weedicide use, preserving sacred groves, and creating institutional herbal gardens in every school and college.

Action Checklist for Medicinal Plant Conservation :-

- ❑ Establish a designated 'Arogya Van' (Medicinal Herbal Garden) on your college campus.
- ❑ Grow a 'Living Herbal First-Aid Box' at home with potted Tulsi, Adulsa, Korphad, and Lemongrass.
- ❑ Document traditional home remedies practiced by grandparents into a personal family ethnobotanical register.
- ❑ Oppose the indiscriminate spraying of chemical weedicides along rural roadsides and field margins.
- ❑ Learn to identify wild medicinal plants and share their conservation value with neighborhood communities.

8. Bibliography / References (संदर्भ सूची)

1. Personal Audio Interview Transcript with Vaidu Shri Ramdas Patil (Conducted August 2026).
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3. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.
4. Kirtikar, K. R., and Basu, B. D. (1935). *Indian Medicinal Plants*. Lalit Mohan Basu, Allahabad.